Figure 4 Walkthrough: Power Heatmaps from Hendrickson et al. (2020)

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Historical reference. This document describes the Figure 4 methodology as implemented in the original Hendrickson et al. (2020) publication code (commit `42ac030`). Several aspects have been superseded by the revised codebase:

- Compound symmetry correlation has been replaced by AR(1) autocorrelation
- The `lme4::lmer` analysis model has been replaced by `nlme::lme` with `corCAR1`
- The scale factor of 2 on carryover decay has been removed
- The step-function BM-BR correlation has been replaced by exponential decay (λ_{cor})
- The parameter grid ($c_{bm} = 0.6$, $\rho = 0.8$) is infeasible under the revised MVN architecture but is now evaluable under Architecture A (mean moderation)

For the revised analysis, see `04-revised-power-analysis.tex`. For the dual-architecture framework, see `02-dgp-mean-moderation-vs-mvn.tex`.

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This document provides a complete technical walkthrough of the power heatmaps in Figure 4 of Hendrickson RC, Thomas RG, Schork NJ, and Raskind MA (2020), 'Optimizing Aggregated N-Of-1 Trial Designs for Predictive Biomarker Validation,' *Frontiers in Digital Health* 2:13. It covers (1) the methodology behind the simulations, (2) the code that implements them in the `pmsimstats` package, and (3) how to reproduce the figures from the bundled data.

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1 Scientific Question

Figure 4 addresses a central question in precision medicine trial design: *What is the statistical power to detect a biomarker-treatment interaction across four candidate trial designs, and how does that power depend on sample size, the strength of the biomarker-response correlation, carryover effects, and participant dropout patterns?*

The biomarker is standing systolic blood pressure; the outcome is PTSD symptom severity (CAPS score); the treatment is the alpha-1 adrenoceptor antagonist prazosin. The interaction of interest is whether baseline blood pressure predicts individual treatment response – not whether the drug works on average, but whether the biomarker identifies who will respond.

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2 The Four Trial Designs

Each design spans 20 weeks with 8 assessment timepoints (excluding baseline). They differ in blinding, randomization structure, and the number of distinct paths a participant can follow through the trial.

2.1 Open-Label (OL) – 1 path

All participants receive active drug for 20 weeks, unblinded. Expectancy is 1.0 at all timepoints. There is no within-subject drug variation, so the analysis tests whether the biomarker predicts the *rate* of improvement over time (`bm:t` interaction).

```
tdOL <- buildtrialdesign(
  name_longform = "open␣label",
  name_shortform = "OL",
  timepoints = cumulative(rep(2.5, 8)),
  timeptname = paste("OL", 1:8, sep = ""),
  expectancies = rep(1, 8),
  ondrug = list(pathA = rep(1, 8))
)
```

2.2 Open-Label + Blinded Discontinuation (OL+BDC) – 2 paths

16 weeks open-label on drug, then a 4-week blinded discontinuation block. In the blinded block, participants are randomized to receive active drug for the first 1 or 2 weeks before switching to placebo. Expectancy drops to 0.5 during the blinded phase.

```
tdOLBDC <- buildtrialdesign(
  name_longform = "open␣label+blinded␣discontinuation",
  name_shortform = "OL+BDC",
  timepoints = c(4, 8, 12, 16, 17, 18, 19, 20),
  timeptname = c("OL1","OL2","OL3","OL4",
                 "BD1","BD2","BD3","BD4"),
  expectancies = c(1, 1, 1, 1, .5, .5, .5, .5),
  ondrug = list(
    pathA = c(1, 1, 1, 1, 1, 1, 0, 0),
    pathB = c(1, 1, 1, 1, 1, 0, 0, 0)
  )
)
```

2.3 Traditional Crossover (CO) – 2 paths

Two 10-week blocks: one on drug, one on placebo, order randomized. All blinded (expectancy = 0.5). No washout period.

```
tdCO <- buildtrialdesign(
  name_longform = "traditional␣crossover",
  name_shortform = "CO",
  timepoints = cumulative(rep(2.5, 8)),
  timeptname = c(paste("COa", 1:4, sep = ""),
                 paste("COb", 1:4, sep = "")),
  expectancies = rep(.5, 8),
```

```
    ondrug = list(
      pathA = c(1, 1, 1, 1, 0, 0, 0, 0),
      pathB = c(0, 0, 0, 0, 1, 1, 1, 1)
    )
  )
```

2.4 N-of-1 (Hybrid) – 4 paths

8-week open-label phase followed by a 4-week blinded discontinuation and two 4-week crossover blocks. Two independent randomization points create 4 possible paths. The timepoints are non-uniform: c(4, 8, 9, 10, 11, 12, 16, 20), with weekly assessments clustered around the blinded-discontinuation transition.

```
  tdNof1 <- buildtrialdesign(
    name_longform = "primary␣N-of-1␣design",
    name_shortform = "N-of-1",
    timepoints = c(4, 8, 9, 10, 11, 12, 16, 20),
    timeptname = c("OL1","OL2","BD1","BD2",
                   "BD3","BD4","COd","COp"),
    expectancies = c(1, 1, .5, .5, .5, .5, .5, .5),
    ondrug = list(
      pathA = c(1, 1, 1, 1, 0, 0, 1, 0),
      pathB = c(1, 1, 1, 1, 0, 0, 0, 1),
      pathC = c(1, 1, 1, 0, 0, 0, 1, 0),
      pathD = c(1, 1, 1, 0, 0, 0, 0, 1)
    )
  )
```

The `buildtrialdesign()` function computes three derived timing variables from the `ondrug` vector:

- **tod** (time on drug): cumulative weeks on active treatment, resetting to 0 when off drug
- **tsd** (time since discontinuation): cumulative weeks off drug since last on-drug period, zeroed before the participant is ever on drug
- **tpb** (time in positive belief): cumulative weeks where expectancy > 0

These drive the Gompertz trajectory calculations in the DGP.

---

3 The Data-Generating Process

3.1 Response Decomposition

The observed outcome (symptom score) at each timepoint t for participant i is:

`Symptom_{i,t} = Baseline_i - [TV_{i,t} + ER_{i,t} + BR_{i,t}]`

Higher factor values represent greater improvement, so they are subtracted from baseline. The three factors are:

- **TV (time-variant response, TR in the paper):** Natural history unrelated to treatment. Includes regression to the mean, the effect of study participation, and time-dependent fluctuations.
- **ER (expectancy-related response):** Response attributable to the participant's belief that they are receiving active treatment. Scaled by the expectancy value (1.0 for open-label, 0.5 for blinded, 0 when expectancy is absent).
- **BR (biologic response):** Direct pharmacologic effect of the drug. This is the only factor that depends on actual drug exposure.

3.2 Modified Gompertz Function

Each factor's expected trajectory is modeled by a modified Gompertz curve:

```
modgompertz <- function(t, maxr, disp, rate) {
  y <- maxr * exp(-disp * exp(-rate * t))
  vert_offset <- maxr * exp(-disp * exp(-rate * 0))
  y <- y - vert_offset
  y <- y * (maxr / (maxr - vert_offset))
  y
}
```

This function guarantees f(0) = 0 and f(infinity) = maxr. The three parameters control:

- `maxr`: asymptotic maximum response
- `disp`: displacement (controls the inflection point)
- `rate`: growth rate

The 'tabula rasa' (TR) parameters extracted from pilot data:

Factor	max	disp	rate	sd
TV	6.506	5	0.35	10
ER	6.506	5	0.35	10
BR	10.986	5	0.42	8

TV and ER share parameters because pilot data did not allow separating them (only the combined placebo response was observed).

3.3 Mean Vector Construction

For a single path with nP = 8 timepoints, the mean vector has 2 + 3*8 = 26 elements:

```
means = [bm_mean, BL_mean,
         TV_1 ... TV_8,
         ER_1 ... ER_8,
         BR_1 ... BR_8]
```

Where:

```
# Baseline (extracted from pilot data):
bm_mean = 124.328 # systolic BP
BL_mean = 83.069  # CAPS score

# Factor means at each timepoint:
TV_t = modgompertz(t_cumulative, 6.506, 5, 0.35)
ER_t = modgompertz(tpb, 6.506, 5, 0.35) * expectancy_t
BR_t = modgompertz(tod, 10.986, 5, 0.42) + carryover_t
```

3.4 Carryover Effect

When a participant transitions from on-drug to off-drug, the BR mean does not immediately drop to zero. Instead, the carryover from the previous timepoint decays exponentially:

```
if (!onDrug[t] && tsd[t] > 0) {
  BR_t = BR_t + BR_{t-1} *
         (1/2)^(scalefactor * tsd[t] / carryover_t1half)
}
```

Key features:

- **Sequential propagation:** Carryover at t depends on the carryover-adjusted value at t-1, not the raw value. This allows residual effects to chain across multiple off-drug periods.
- **Scale factor = 2** (default): The drug effect decays faster than the raw half-life would suggest, reflecting the combined effect of pharmacokinetic clearance and clinical re-emergence of symptoms.
- **Half-life values tested:** 0, 0.1, 0.2 weeks. Even 0.1 weeks (~17 hours) produces substantial power loss.
- **BR only:** Carryover is not applied to TV or ER.

3.5 Covariance Matrix

The 26x26 covariance matrix encodes correlations between all variables. It is constructed from a correlation matrix and a vector of standard deviations:

```
sigma <- outer(sds, sds) * correlations
```

Standard deviations:

```
sds = [bm_sd, BL_sd,
       rep(TV_sd, 8),
       rep(ER_sd, 8) * expectancy,
       rep(BR_sd, 8)]
```

Where `bm_sd = 15.362`, `BL_sd = 18.483`, `TV_sd = 10`, `ER_sd = 10`, `BR_sd = 8`. The ER standard deviations are scaled by the expectancy value at each timepoint (0 expectancy means 0 variance in ER).

Correlation structure (fixed across all simulations):

Correlation	Value	Meaning
c.tv, c.pb, c.br	0.8	Within-factor autocorrelation (compound symmetry)
c.cf1t	0.2	Cross-factor correlation at the same timepoint
c.cfct	0.1	Cross-factor correlation at different timepoints

Biomarker-BR correlation (the critical interaction mechanism):

The correlation between the biomarker (`bm`) and the bio-response (`BR_t`) at each timepoint determines the biomarker-treatment interaction. This is set according to Ron Thomas's logic:

- **Timepoint 1:** Always 0 (no drug effect at first measurement)
- **On-drug timepoints (raw BR > 0):** `Cor(bm, BR_t) = c.bm`
- **Off-drug, no carryover residual (BR $= 0$):** `Cor(bm, BR_t) = 0`
- **Off-drug, with carryover residual (raw BR was 0 but adjusted BR > 0):** `Cor(bm, BR_t) = (BR_t / BR_{t-1}) * c.bm` (scaled proportionally to the carryover fraction)

This differential correlation IS the biomarker-treatment interaction. There is no population-level mean shift; the interaction emerges from the fact that participants with higher biomarker values have BR draws that are more correlated (when on drug) or less correlated (when off drug) with the biomarker.

Positive definiteness: After construction, the covariance matrix is checked via `corpcor::is.positive.definite(` If it fails, it is forced positive definite via `corpcor::make.positive.definite(sigma, tol = 1e-3)`.

3.6 Multivariate Normal Draw

With the mean vector and covariance matrix in hand, N participants are drawn from a multivariate normal distribution:

```
dat <- MASS::mvrnorm(n = N, mu = means, Sigma = sigma)
```

This produces a data.table where each row is a participant and the columns are: biomarker, baseline, and 24 factor values (3 factors x 8 timepoints). The outcome at each timepoint is then computed as:

`Symptom_t = BL - (TV_t + ER_t + BR_t)`

This is done via `eval(parse(text = ...))` in `generateData.R` lines 200–215, constructing column expressions dynamically.

---

4 Censoring (Dropout) Model

The censoring model simulates realistic participant dropout with two components:

1. **Time-dependent (flat) dropout:** Probability increases linearly with the duration of each interval. Controlled by `beta0 * (1 - beta1)`.

2. **Outcome-dependent (biased) dropout:** Probability is higher for participants whose symptoms have not improved (or have worsened). Computed as `(symptom_change + 100)^eb1`, where the shift by 100 ensures all values are positive. Controlled by `beta0 * beta1`.

Once a timepoint is censored, all subsequent timepoints for that participant are also set to NA (last-observation-carried-forward censoring).

Censoring parameter sets used in Figure 4:

Name	beta0	beta1	eb1	Description
No censoring	–	–	–	Complete data
balanced	0.05	0.5	2	Equal flat and biased dropout
more of flat	0.05	0.2	2	Mostly time-dependent dropout
more of biased	0.05	0.8	2	Mostly outcome-dependent dropout
high dropout	0.15	0.5	2	Triple the overall dropout rate

---

5 The Analysis Model

Each simulated trial is analyzed using a linear mixed-effects model (`lme4::lmer`). The model formula is selected dynamically based on the trial design structure.

5.1 Designs with within-subject drug variation (CO, N-of-1, OL+BDC)

The formula is:

```
Sx ~ bm + t + Dbc + bm:Dbc + (1|ptID)
```

Where:

- `Sx`: symptom score at each timepoint
- `bm`: baseline biomarker value (continuous)
- `t`: time since baseline (continuous)
- `Dbc`: continuous drug indicator. When on drug: `Dbc = 1`. When off drug with carryover modeling: `Dbc = (1/2)^(scalefactor * tsd / carryover_t1half)`. When off drug without carryover: `Dbc = 0`.
- `bm:Dbc`: the biomarker-treatment interaction (primary outcome of interest)
- `(1|ptID)`: random intercept per participant

The coefficient of `bm:Dbc` represents the extent to which the biomarker predicts differential treatment response. A significant negative coefficient indicates that higher biomarker values predict greater symptom improvement when on drug.

5.2 Designs without within-subject drug variation (OL)

When all participants are always on drug, there is no variation in `Dbc`, so the model cannot estimate `bm:Dbc`. Instead:

```
Sx ~ bm + t + bm:t + (1|ptID)
```

The `bm:t` coefficient tests whether the biomarker predicts the rate of improvement over time.

5.3 Model construction in code

The formula is built as a string and evaluated:

```r
# lme_analysis.R lines 133-151
modelbase <- "Sx~bm+t"
if (varInDb) {
  modelbase <- paste0(modelbase, "+Dbc+bm*Dbc")
} else {
  modelbase <- paste0(modelbase, "+bm*t")
}
# Optional terms for expectancy, carryover, random slopes...
modelbase <- paste0(modelbase, "+(1|ptID)")
eval(parse(text = paste0("form<-", modelbase)))
fit <- lmer(form, data = datamerged)
```

5.4 Output extraction

The p-value for the interaction term is extracted from the `lmerTest` summary (which uses Satterthwaite degrees of freedom):

```r
c <- summary(fit)$coefficients
if (varInDb) {
  p    <- c["bm:Dbc", "Pr(>|t|)"]
  beta <- c["bm:Dbc", "Estimate"]
  betaSE <- c["bm:Dbc", "Std.␣Error"]
} else {
  p    <- c["bm:t", "Pr(>|t|)"]
  beta <- c["bm:t", "Estimate"]
  betaSE <- c["bm:t", "Std.␣Error"]
}
```

---

6 The Simulation Loop

`generateSimulatedResults()` orchestrates the parameter sweep:

1. **Expand the parameter grid:** `expand.grid(trialdesign, respparamset, blparamset, modelparamset)` – note that censoring is applied after data generation, not included in the grid.
2. **For each parameter combination:**

- Generate `N * Nreps` participants at once (for efficiency) across all paths
- Split into `Nreps` replicates
- Fit the LME model to the full population to get the 'empirical true beta' (ETbeta)
- Fit the LME model to each replicate to get per-replicate beta, betaSE, p-value
- Optionally apply each censoring pattern and re-analyze

3. **Output:** A data.table with one row per (parameter-combination x replicate x censoring-pattern), containing: beta, betaSE, p, issingular, warning, ETbeta, ETbetaSE, frac_NA.

6.1 Model parameters for Figure 4

```
coremodelparams <- expand.grid(
  N = c(35, 70),
  c.bm = c(0, 0.3, 0.6),
  carryover_t1half = c(0, 0.1, 0.2),
  c.tv = 0.8, c.pb = 0.8, c.br = 0.8,
  c.cf1t = 0.2, c.cfct = 0.1
)
```

This gives 2 x 3 x 3 = 18 model parameter combinations, times 4 trial designs, times 1 response parameter set, times 1 baseline parameter set = 72 cells in the design grid. Each cell is replicated 1,000 times. Each replicate is analyzed with and without each of 4 censoring patterns, yielding 5 analysis conditions per replicate.

Analysis parameters: `useDE = FALSE` (expectancy not included as a fixed effect), `t_random_slope = FALSE` (random intercept only), `full_model_out = FALSE`.

---

7 Figure 4 Heatmap Construction

7.1 Figure 4A: Effect of trial design, censoring, and biomarker effect on power

```
param2vary <- c("trialdesign", "N", "c.bm",
               "censorparamset")
param2hold <- data.table(blparamset = 1,
                        respparamset = 1,
                        carryover_t1half = 0)
param2nothold <- c("c.cfct", "modelparamset")
param2plot <- "power"
```

7.2 Figure 4B: Effect of nonzero carryover term on power

```
param2vary <- c("trialdesign", "N",
               "carryover_t1half",
               "censorparamset")
param2hold <- data.table(blparamset = 1,
                        respparamset = 1,
```

Mapping	Parameter	Values
Facet columns	trialdesign	OL, OL+BDC, CO, N-of-1
Facet rows (right axis)	N	35, 70
X-axis	c.bm	0, 0.3, 0.6
Y-axis	censorparamset	No censoring, balanced, more of flat, more of biased, high dropout
Fill color	power	0.0 to 1.0
Held constant	carryover_t1half	0

```
                        c.bm = 0.6)
param2nothold <- c("c.cfct", "modelparamset")
param2plot <- "power"
```

Mapping	Parameter	Values
Facet columns	trialdesign	OL, OL+BDC, CO, N-of-1
Facet rows (right axis)	N	35, 70
X-axis	carryover_t1half	0, 0.1, 0.2
Y-axis	censorparamset	No censoring, balanced, more of flat, more of biased, high dropout
Fill color	power	0.0 to 1.0
Held constant	c.bm	0.6

7.3 How `PlotModelingResults()` computes power

The function (plottingfunctions.R:26–191) operates as follows:

1. **Filter** `data$results` to the held-constant parameters. For parameters not explicitly held or varied, the function takes the first available value (line 92–94).

2. **Convert facet parameters** (first two in `param2vary`) to factors with human-readable labels from the trial design metadata and censoring parameter names.

3. **Compute summary statistics** grouped by the four varying parameters:

   ```
   d[, sig_p := (p < 0.05), by = tID]
   d[, power := mean(sig_p), by = op]
   ```

   Power is the fraction of replicates where the interaction term is significant at alpha = 0.05.

4. **Plot** using `ggplot2`:

   ```
   ggplot(data = ds, aes_string(op[3], op[4])) +
     geom_tile(aes(fill = toplot), colour = "white") +
     scale_fill_distiller(type = "seq", palette = "RdYlBu") +
     geom_text(aes(label = round(toplot, 2)), size = 3) +
     facet_grid(reformulate(op[1], op[2]))
   ```

   The RdYlBu palette maps low power (red) through yellow to high power (blue).

8 Reproduction Script

The bundled `results_core.rda` dataset contains the pre-computed simulation results (1,000 replicates). To reproduce Figure 4:

```
library(pmsimstats)
library(data.table)
library(ggplot2)
library(gridExtra)

data(results_core)
simresults <- results_core

# --- Figure 4A ---
p1 <- PlotModelingResults(
  simresults,
  param2plot = "power",
  param2vary = c("trialdesign", "N", "c.bm",
                 "censorparamset"),
  param2hold = data.table(blparamset = 1,
                          respparamset = 1,
                          carryover_t1half = 0),
  param2nothold = c("c.cfct", "modelparamset")
)

# --- Figure 4B ---
p2 <- PlotModelingResults(
  simresults,
  param2plot = "power",
  param2vary = c("trialdesign", "N",
                 "carryover_t1half",
                 "censorparamset"),
  param2hold = data.table(blparamset = 1,
                          respparamset = 1,
                          c.bm = 0.6),
  param2nothold = c("c.cfct", "modelparamset")
)

# --- Compose ---
p1 <- p1 +
  ggtitle(paste("A. Effect of trial design, censoring",
                "and biomarker effect on power")) +
  theme(plot.title = element_text(hjust = 0))

p2 <- p2 +
  ggtitle(paste("B. Effect of nonzero carryover term",
                "on power")) +
  theme(plot.title = element_text(hjust = 0))

grid.arrange(p1, p2, nrow = 2)
```

To re-run the simulations from scratch (rather than using the bundled data), set `Nreps <- 1000` in the first vignette and run the `coremodelparams` block. On a modern machine, expect approximately 8–12 hours for the full parameter space.

9 Code Map

Step	File	Lines	Function
Define trial designs	buildtrialdesign.R	75–119	`buildtrialdesign()`
Compute tod/tsd/tpb	buildtrialdesign.R	88–106	(within `buildtrialdesign()`)
Gompertz trajectories	utilities.R	34–40	`modgompertz()`
Cumulative time helper	utilities.R	8–18	`cumulative()`
Build mean vector	generateData.R	63–103	(within `generateData()`)
Carryover on BR means	generateData.R	86–100	(within `generateData()`)
Build correlation matrix	generateData.R	105–163	(within `generateData()`)
BM-BR correlation logic	generateData.R	146–162	Ron Thomas scaling
Covariance from correlation	generateData.R	172	`outer(sds, sds) * correlations`
PD enforcement	generateData.R	177–181	`corpcor::make.positive.definite()`
MVN draw	generateData.R	188	`MASS::mvrnorm()`
Compute outcomes	generateData.R	200–215	`eval(parse())` column construction
Censoring/dropout	censordata.R	39–89	`censordata()`
Parameter sweep loop	generateSimulatedResults.R	66–255	`generateSimulatedResults()`
LME model construction	lme_analysis.R	133–151	Dynamic formula via `eval(parse())`
Interaction extraction	lme_analysis.R	196–205	`bm:Dbc` or `bm:t` coefficient
Heatmap visualization	plottingfunctions.R	26–191	`PlotModelingResults()`
Trajectory visualization	plottingfunctions.R	220–338	`plotfactortrajectories()`

10 Key Findings from Figure 4

10.1 Figure 4A (no carryover)

- **N-of-1 has the highest power** across all conditions, reaching 0.95 at N=35, c.bm=0.6 without censoring.
- **OL has the lowest power** because it lacks within-subject drug variation and can only detect the biomarker-time interaction.
- **CO has slightly lower power than N-of-1** despite full blinding, because the N-of-1 design includes both open-label and crossover information.
- **Censoring reduces power** across all designs, with the strongest effect on OL+BDC (which has the shortest blinded phase, making it most vulnerable to pre-blinding dropout).
- **c.bm = 0 produces power near 0.05** across all designs, confirming appropriate Type I error control.

10.2 Figure 4B (carryover at c.bm = 0.6)

- **Even short carryover (0.1 weeks) precipitously reduces power** for N-of-1 and OL+BDC designs. N-of-1 drops from 0.95 to ~0.18 at N=35.

- **CO is least affected** because it has only one transition from on-drug to off-drug (or vice versa), while N-of-1 has multiple transitions amplifying carryover contamination.
- **OL is unaffected** by carryover because participants are always on drug.
- The vulnerability of the N-of-1 design to carryover is the paper's most clinically important finding: the design that offers the best power under ideal conditions is also the most sensitive to even minimal carryover effects.